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17 April 1967
SUPERSEDING
FA-PD-MI-2660, Rev. 0
30 April 1965

MILITARY SPECIFICATION
PROJECTILE, AP-T, 90MM, M318A1
METAL PARTS

1. SCOPE

1.1 This specification covers Projectile, AP-T, 90MM, M318A1, Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

MIL-M-11472 - Magnetic Particle Inspection, Process for Ferromagnetic Materials

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing

Federal Test Method Standard No. 151 - Metals: Test Method

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-410 - Certification of Penetrant Inspection Personnel

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1169 - Packaging, Packing and Marking for inert ammunition Components

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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2.1 (Cont'd)

DRAWINGS

Munitions Command

F10522523	- Projectile, 90MM, AP-T M318A1 Metal Parts
B8597489	- Plug and Disc Assembly
F7548604	- Pallet, for Fixed and Semifixed Projectile, Packaging and Marking
IEL10522523	- Index of Inspection Equipment List for Projectile, AP-T, 90MM, M318A1, Metal Parts
D8593429	- Band seating testing machine indentor and test data.

PUBLICATIONS

Munitions Command

Operating Instructions for Use of Detroit Testing Machine for Band Seating Inspection	
PA-PD-792	- Equipment, Testing or Measuring for Ammunition Items

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA - B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirement specified on the applicable drawings and referenced specifications.

3.2 Metal Defects.

3.2.1 Projectile Body.-The machined, heat treated body shall be free of cracks, seams, inclusions and surface defects which may have a deleterious effect in ballistics performance of the end item. Surface imperfections (other than cracks) and nonmetallic inclusions greater than 1/8 inch in length on the ogive surface and greater than 2 inches in length on the cylindrical surface of the body are not permissible.

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3.2.2 Windshield.-Fusion defects such as porosity, cold shuts, shrinkage, cracks, etc. shall be limited by acceptable radiographic standards representative of good foundry control.

3.3 Mechanical Properties.

3.3.1 Projectile Body.-Mechanical properties and hardness shall be compatible with required terminal ballistics capability. Hardness pattern shall be established and maintained throughout production which will assure consistent defeat of required armor thickness, (see figure 1).

3.3.2 Rotating Band.-After seating and subsequent removal from the projectile body, the rotating band shall have the following mechanical properties:

Tensile strength - 32,000 pounds per square inch - minimum.
Elongation - 15 percent per inch of gage length - minimum.

3.4 Engineering Data.-An engineering data sheet shall be furnished with each lot and forwarded to the Proving Ground on a separate card. This card shall contain the following information:

a. Ladle analysis of each heat of steel.
b. Heat treatment.
c. Hardness pattern through the longitudinal section from the tip to the base for one projectile body from each heat of steel.

3.5 Application of Rotating Band.

3.5.1 Band Seat.-Prior to application of the rotating band blank, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter.

3.5.2 Band Blank (Gilding Metal).-The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification.

3.5.3 Application.-The rotating band blank shall be applied cold and shall be carefully pressed into the band seat so as to make contact with the sidewalls for a minimum of 8/10 of the band seat depth.

3.5.4 Tightness.-The average indicated diametrical clearance between the rotating band and band seat shall not exceed .010 inch.

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3.6 Functional

3.6.1 Penetration.-The protective ballistic limit (PBL) of the as-fired projectile shall not exceed the PBL of the reference projectile by more than 150 feet per second (fps), when fired against 3-inch thick homogeneous armorplate oriented at an angle of 55 degrees with the vertical.

3.6.2 Security.-There shall be no loss or breakup of any projectile component either in the gun bore or in flight, when fired with a propelling charge adjusted to produce 115% of the service pressure of the round in a new gun.

3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Surface Preparation.-Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.7.2 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment.

3.7.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.7.4 Rotating Band.-The machined rotating band shall not be nicked, or otherwise damaged in subsequent handling prior to shipment.

3.7.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.7.6 Foreign Matter.-No part or assembly shall contain dirt, grease, chips, rust or other foreign matter. Discoloration not due to corrosive action will be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial

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laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550 except as specified herein.

4.2 First Article Sample

4.2.1 Initial Production Sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 44 projectile metal parts as follows:

34 Projectile metal parts complete (painted)
5 Projectile metal parts (unpainted)
5 Sets of component parts unassembled including rotating band blanks

Twenty-nine of the above projectile metal parts (painted) shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government Laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval

4.3 Inspection Provisions.

4.3.1 Lot.

4.3.1.1 Submission of Product.-The supplier shall submit the product in accordance with MIL-STD-105.

4.3.1.1.1 Lot Numbering.-Lot numbers shall be assigned in accordance with the criteria of MIL-STD-1167.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Windshield, Drawing C10522526 covering a detail of Drawing F10522523.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of radius in base, with large outside diameter	Gage
102.	Diameter outside, large	Gage
103.	Location of small inside basic diameter	Gage
104.	Location of large inside basic diameter	Gage
105.	Length of ogive mating surface	Gage
106.	Location of small outside basic diameter	Gage
107.	Location of large outside basic diameter	Gage
108.	Wall thickness	Gage
109.	Metal defective	Visual
MINOR		
201.	Length total	Gage
202.	Surface finish improper	Visual (see 6.2)
203.	Burr	Visual
204.	Foreign matter	Visual

4.3.2.2 Pin Assembly Drawing B8597486 (if applicable) covering a detail of Drawing F10522523

CRITICAL		
	None defined	
MAJOR		
101.	Diameter of spring pin, min.	Gage
102.	Length of spring pin	Gage
103.	Diameter of rivet head	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Improper assembly	Visual/Manual
202. Rivet binding completely	Visual/Manual
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.3 Body, Drawing F10522525, covering a detail of Drawing F10522524.

MAJOR	
101. Diameter of rotating band seat	Gage
102. Width of rotating band seat	Gage
103. Location of rotation band seat	Gage
104. Foreign matter on rotation band seat	Visual
MINOR	
201. Fillet(s) missing from rotating band seat	Visual (see 6.2)
202. Scoring missing from rotating band seat	Visual
203. Rib(s) missing from rotating band seat	Visual

4.3.2.4 Body Assembly, Drawing D10522524 covering a detail of Drawing F10522523.

CRITICAL
None defined

MAJOR	
101. Concentricity of nose with bourrelet	Gage
102. Location of small basic diameter of ogive	Gage
103. Location of large basic diameter of ogive	Gage
MINOR	
201. Width of nose flat, max. (if applicable)	Over Screen Chart

4.3.2.5 Metal Parts, Drawing F10522523 (prior to painting and assembly of spring pins.)

CRITICAL
None defined

MAJOR	
101. Diameter of spring pin hole	Gage
102. Depth of spring pin hole (max.)	Gage
103. Location of spring pin holes	Gage

MINOR
None defined

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4.3.2.6 Metal Parts, Drawing F10522523 (prior to painting and application of filler).

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of windshield with bourrelet	Gage
102.	Concentricity of rotating band with bourrelet	Gage
103.	Concentricity of body with bourrelet	Gage
104.	Concentricity of tracer cavity with bourrelet	Gage
105.	Diameter of bourrelet, min.	Gage
106.	Diameter of rotating band	Gage
107.	Location of rotating band	Gage
108.	Angle at front of rotating band	Gage
109.	Location of angle of rotating band	Gage
110.	Width of rotating band	Gage
111.	Diameter of rotating band relief groove	Gage
112.	Angle at rotating band relief groove incorrect	Gage
113.	Location of crimping groove	Gage
114.	Depth of crimping groove	Gage
115.	Radius of crimping groove	Gage
116.	Epoxy not visible for full 360 degrees	Visual
117.	Epoxy protruding beyond bourrelet diameter	Visual
118.	Evidence of grinding rear of windshield after removal of Epoxy	Visual
119.	Magnetic particle stamp missing (indicating failure to comply with requirements)	Visual
120.	Spring pins missing or loose	Visual/Manual
MINOR		
201.	Diameter of body	Gage
202.	Location of front bourrelet	Gage
203.	Location of rear bourrelet	Gage
204.	Depth of tracer cavity	Gage
205.	Depth of tracer cavity drill point, max.	Gage
206.	Location of angle at rear of body, min.	Gage
207.	Depth of undercut front and rear of rotating band, max.	Visual (see 6.2)
208.	Width of undercut front and rear of rotating band, max.	Visual (see 6.2)
209.	Radii or chamfers missing or incorrect	Visual (see 6.2)
210.	Surface finish, improper	Visual (see 6.2)
211.	Spring pins protruding beyond surface of windshield	Visual
212.	Rivet protruding beyond surface of windshield	Visual
213.	Marking, missing, incorrect or illegible	Visual
214.	Burr	Visual
215.	Foreign matter	Visual

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4.3.2.7 Metal Parts, Drawing F10522523 (after painting and application of filler but prior to insertion of dessicant).

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Weight	Approved Scale
102.	Tracer cavity not functionally concentric with minor diameter of thread	Gage
103.	Pitch diameter of tracer thread, max.	Gage
104.	Minor diameter of tracer thread, max.	Gage
105.	Diameter of bourrelet, max.	Gage
106.	Diameter of tracer cavity	Gage
107.	Length of effective tracer thread, min	Gage
108.	Length total, max.	Gage
109.	Insecure assembly	Visual/Manual
110.	Tracer thread damaged or not full	Visual
111.	Rotating band damaged	Visual
MINOR		
201.	Pitch diameter of tracer thread, min.	Gage
202.	Presence of chamfer at tracer cavity	Visual (see 6.2)
203.	Presence of tracer thread undercut (if applicable)	Visual (see 6.2)
204.	Protective coating inadequate or damaged	Visual
205.	Paint or rotating band bearing surfaces	Visual
206.	Paint in tracer cavity including threads	Visual
207.	Filler covering spring pins, above windshield surface	Visual/Tactile
208.	Burr	Visual
209.	Foreign matter	Visual

4.3.2.8 Metal Parts, Drawing F10522523 (after insertion of dessicant).

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Dessicant missing from tracer cavity	Visual
202.	Tracer cavity not properly sealed	Visual

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4.3.2.9 Plug and Disc Assembly, Drawing B8597489.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Inside diameter not functionally concentric with pitch diameter of thread	Gage
102. Pitch diameter of thread, min.	Gage
103. Major diameter of thread, min.	Gage
104. Diameter of hole	Gage
105. Thickness of plug	Gage
106. Thickness of disc	Gage
107. Threads damaged or not full	Visual
108. Improper assembly	Visual
109. Metal defective	Visual
110. Protective coating, inadequate or damaged	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Distance between wrench holes	Gage
203. Surface finish, improper	Visual (see 6.2)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.10 Carton (prior to sealing) (Plug and Disc Assembly).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Support missing or damaged	Visual
203. Separator missing or damaged	Visual
204. Incorrect number in carton	Visual

4.3.2.11 Carton (after sealing) (Plug and Disc Assembly).

CRITICAL
None defined

MAJOR
None defined

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not properly sealed	Visual
203. Contents loose	Manual
204. Marking missing, incorrect or illegible	Visual

4.3.2.12 Pallet, Drawing F7548604 (Projectile).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve(s) missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual/Manual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection Testing.-The following tests shall be performed in accordance with 4.4:

4.3.3.1 Magnetic Particle (Major Defect).-Each projectile body shall be subjected to a magnetic particle test to determine compliance with 3.2.1. Failure of any body to comply shall be cause for rejection of the body.

4.3.3.2 Mechanical Properties.

4.3.3.2.1 Body.-Hardness Pattern - One projectile body shall be selected from each heat of steel in the lot, be used to develop the hardness pattern required by 3.4.c.

4.3.3.2.2 Rotating Band.-One projectile body assembly, after banding but prior to machining of the rotating band, shall be selected from each 8-hour production or fraction thereof for each banding machine in operation to determine compliance with 3.3.2. Failure of the band to comply shall be cause for rejection of the production represented by the sample.

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4.3.3.3 Band Seating.

4.3.3.3.1 Tightness.-Ten projectiles from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test to determine compliance with 3.5.4. Failure to comply shall be cause for rejection of the production processed since the last acceptable test was performed.

4.3.3.3.2 Sidewall Contact.-Rotating bands removed as specified under 4.3.3.2.2 shall be examined to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of body assemblies represented by the sample.

4.3.3.4 Protective Coating.-The sample size and acceptance criteria for the inspection of the cleaning and the application of the protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.3.5 Assembly (Major Defect).-Each shot assembly shall be visually examined to determine compliance with the applicable assembly drawing (10522325). Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.4 Functional.-The following functional tests will be performed by the Government:

TABLE I

Test Phase	Sample Size	
	Initial Production Sample	Production Sample
Penetration	9	9
Security	10	5
Velocity-Accuracy	10	5

4.3.4.1 Penetration.-The samples selected in accordance with Table I will be subjected to this test, to determine compliance with 3.6.1. Failure of any projectile to comply will be cause for rejection of the lot.

4.3.4.2 Security.-The samples selected in accordance with Table I, for this phase will be subjected to this test to determine compliance with 3.6.2. Failure of any projectile to comply will be cause for rejection of the lot.

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4.3.4.3 Velocity-Accuracy.-The samples selected in accordance with Table I for this phase will be subjected to this test for information only, except that failure of any projectile in this sample to comply with 3.6.2 will be cause for rejection of the lot.

4.3.5 Inspection Equipment.-Index of inspection equipment list IEL 10522523 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test Methods and Procedures.

4.4.1 Magnetic particle (major defect).-The procedures for performing this test shall be as specified in MIL-M-11472, for direct magnetization by the residual method. The suspension shall be tested at the start of each days production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by MIL-M-11472. After performing the test for solid content of suspension, if it is determined that it does not comply, with 3.7.3.2 of MIL-M-11472, all bodies that have been accepted since the last satisfactory test was performed, shall be re-inspected after the suspension has been brought up to the proper level. A magnetizing current of 1200 ± 50 amperes is required. Either direct current (DC) or alternating current (AC) may be used in magnetizing the sample under examination. When using the A.C. method, the current should be cut off at the first quarter (90 degrees) of the A.C. sine-wave to insure that the body is magnetized after acceptance. The body shall be stamped with the letter "M" to signify compliance with 3.2.1. The magnetic particle inspection shall be performed on equipment that complies with the requirements of Specification PA-PD-792. The operator(s) of the equipment and the inspector(s) making the determination shall be certified in accordance with MIL-STD-410.

4.4.2 Mechanical Properties.

4.4.2.1 Body Hardness Pattern.-The sample provided by 4.3.3.2.1 shall be sectionized by removing a full length and width specimen approximately 1-inch thick from middiameter of the projectile. The hardness pattern required by 3.4 shall be developed utilizing the Rockwell "C" scale (diamond brail). The equipment and procedures for performing this test shall be as prescribed in Federal Test Method Standard No. 151.

4.4.2.2 Rotating Band.-The samples selected in accordance with 4.3.3.2.2 shall have the unmachined rotating band removed from the body and carefully flattened without hammering. The tension test specimen shall be cut along the direction of the flattened length and prepared in such a manner as to hold to a minimum any additional cold working. The sample shall be tested in accordance with the criteria for performing the tension test as prescribed by Federal Test Method Standard No. 151.

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4.4.3 Band Seating.

4.4.3.1 Tightness.-The samples provided in 4.3.3.3.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing machine manual (see 6.1) and line 12 of Drawing D8593429, Revision D.

4.4.3.2 Sidewall Contact.-The sample provided by 4.3.3.3.2 shall be examined for fill of the band seat undercut. Care must be exercised in removal of the band to prevent deformation of material which was swaged into the band seat undercut.

4.4.4 Protective Coating.-The procedures for cleaning the components and assemblies and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification, except that the salt spray prescribed shall be performed as specified on the applicable drawing, utilizing the equipment in compliance with Federal Test Method Standard No. 141.

4.4.5 Functional

4.4.5.1 Penetration.-Prior to firing penetration samples, the protective ballistic limit (PBL) of a reserve reference lot of the same caliber and model will be determined. The PBL is defined as the mean of 6 velocities - to include 3 complete and 3 partial penetrations, the velocity spread of which shall not exceed 150 fps. Following determination of the PBL of the reference lot, the PBL of the test lot shall be determined as indicated above utilizing samples selected in accordance with Table I. Sample projectiles will be assembled into complete rounds utilizing the service charge. The rounds will be fired in a gun for which it is standard, against 3-inch thick homogenious armorplate placed approximately 300 ft. from the muzzle of the gun and at an obliquity of 55 degrees. A complete penetration will be considered to have occurred if a fragment of the plate or projectile is thrown beyond the rear face of the plate with sufficient energy to penetrate an 0.020-inch aluminum alloy (24st) sheet, or its equivalent, placed parallel to and one foot behind the plate. When, in the proof officer's opinion, it is possible to observe that these conditions are being complied with without the use of a sheet, the sheet may be omitted. An impact which occurs under the following conditions will be disregarded and another sample fired in its place:

a. Any projectile which strikes within 2 calibers of the edge of the plate.

b. Any projectile which strikes within 2 calibers of a previous impact or whose path of penetration intersects a previous hole.

c. Any projectile which strikes the spalled area of a previous impact.

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4.4.5.2 Security.-The samples provided by Table I assembled into complete rounds will be fired in a gun for which the round is standard. The propellant charge will be adjusted to obtain 115 percent of the maximum rated pressure of the round in a new gun. The projectile will be fired through 2 canvass or heavy muslin screens placed approximately 2000 meters from the muzzle of the gun. Observation will be made for evidence of loss or breakup of any projectile component, either in the gun bore or in flight. A smear camera located near the muzzle will be used to verify security on the initial production sample and on the first five production lots. Prior to firing, paint will be applied to the windshield. Observation for paint on the cloth target will be made for windshield security. Lot acceptance will be based on observation at the 2000 meter range.

4.4.5.3 Velocity-Accuracy.-The samples provided by Table I assembled into complete rounds will be fired in a gun for which the round is standard, utilizing the service charge. The projectile will be fired to impact on a vertical target placed 2000 meters from the muzzle of the gun. The test rounds will be fired alternately with a similar group of reference rounds for comparison of velocity-accuracy at this range. Accuracy in terms of maximum spread in inches, both horizontal and vertical and probable error (PE) in MILS, will be recorded for all rounds fired and forwarded to the cognizant technical agency for evaluation (See 6.3).

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.

5.1.1 Level C. - Packing and marking shall be in accordance with Drawing F7548604 and MIL-STD-1169.

5.1.2 Level C. - Plug and disc assembly, Drawing B8597489 shall be packed in accordance with MIL-STD-1169. Pack 2000 plug and disc assemblies in 2 layers arranged in a 10 x 10 pattern, 10 to a cell, using interlocking separators.

5.2 Data Cards.- Data card information shall be as specified in MIL-STD-1168.

6. NOTES

6.1 Band Seating.-The operating procedures for use of the Detroit testing machine, as prepared by the Inspection Engineering Office, Frankford Arsenal, SMUFA-Q3000, may be used as an aid in checking band seating inspection procedures.

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6.2 Visual Examination Qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Ballistic Test.-Rounds fired for velocity accuracy are not acceptance tests, but the results will be carefully analyzed by the Munitions Command to determine quality of ammunition produced. Evidence of excessive variation in any phase of the Proving Ground tests will result in immediate investigation by the Munitions Command.

6.4 Heat Treatment.-Hardness pattern of figure 1 is advisory and is representative of shot bodies which have satisfied the plate penetration requirements of 3.6.1.

6.4.1 4160 steel with the following chemistry and heat treatment has been used in past production to obtain advisory hardness pattern shown on figure 1.

Chemistry

C = 0.61	S = 0.035	Cr = 1.00
Mn = 0.91	Si = 0.28	Mo = 0.21
Al = 0.017	Ni = 0.00	Va = 0.00

Heat Treatment

1650 degrees F in salt, total time 49 minutes

Quench in salt, 375 degrees F 9 minutes. Air cool to room temperature Temper 350 degrees F in oil 3 hours. Base draw 1250 degrees F, 9 minutes in salt base immersed 3 inches deep air cool to room temperature; nose immersed in water to bourrelet.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 1315-A611

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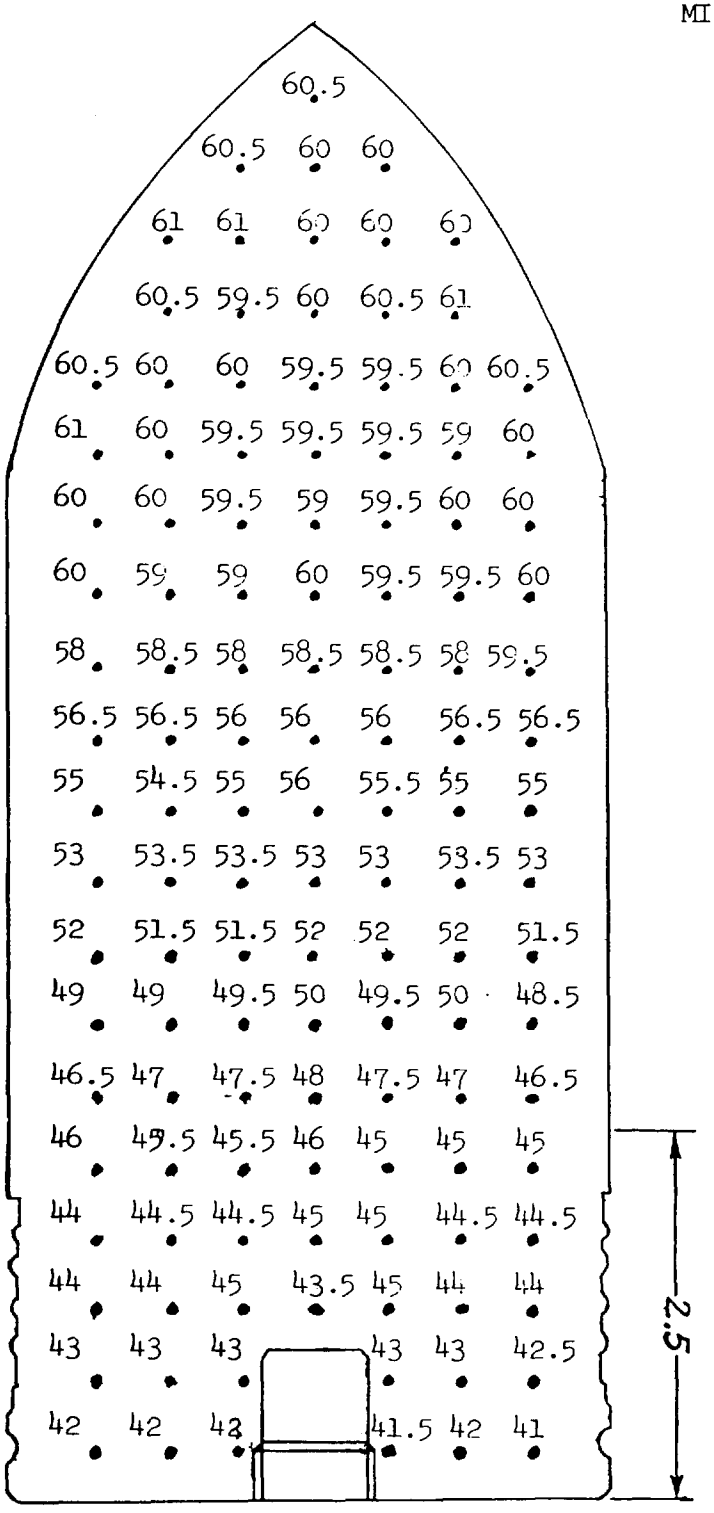


Figure 1: Advisory Heat Treatment Pattern for 90MM, M318A1 AP-T Shot

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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